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E DELL'INFORMAZIONE

Space Mission Analysis Laboratory

FINAL REPORT - AEROSPACE ENGINEERING DEGREE PROGRAM

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1 Introduction

This report presents the final project completed for the Space Mission Analysis Laboratory within the Aerospace Engineering program at Politecnico di Milano. Mission analysis is the engineering discipline concerned with defining and evaluating a space mission by examining its objectives, requirements, and operational constraints in order to assess its feasibility and performance.
This study presents and analyzes a mission to reach the near-Earth asteroid 5797 *Bivoj*, with the aim of assessing its feasibility and key operational characteristics.

The mission analysis was conducted under a set of simplifying assumptions that provide a consistent reference framework for the preliminary mission design.
In particular, the adopted assumptions concern:

- *Classical mechanics* and *Newton's law of universal gravitation.*
- *Two-body approximation*: the motion of the spacecraft is described by assuming that the only significant gravitational interaction is with the central body, while perturbations due to other bodies are neglected.
- *Impulsive maneuvers*: each maneuver is modeled as an instantaneous change in velocity, with zero burn duration and finite Δv.
- *Patched-conics approximation*: the trajectory is divided into conic arcs, each governed exclusively by the gravitational attraction of the relevant central body. The arcs are connected at the corresponding sphere-of-influence boundaries.

Five classical orbital elements define the geometry and orientation of an orbit: the semi-major axis a, eccentricity e, inclination i, right ascension of the ascending node Ω, and argument of periapsis ω. The sixth element, the true anomaly θ, specifies the instantaneous position along the orbit.

The objective is to optimize the mission by identifying an effective compromise between transfer time, propellant consumption, and overall mission cost. The resulting trajectory must remain feasible while satisfying the relevant operational constraints and mission requirements. Particular attention is devoted to minimizing the required Δv, as it directly affects the propellant budget and the overall performance of the mission. The mission analysis is divided into three sections::

- *Section* 1: definition of the maneuvers required to reach the parking orbit;
- *Section* 2: definition of the direct Earth–Bivoj transfer orbit;
- *Section* 3: definition of the Earth-departure and asteroid-arrival hyperbolic trajectories.

The proposed methodology is general and is suitable for the preliminary design of missions aimed at reaching any asteroid. The project was implemented in MATLAB for numerical computation and simulation. The corresponding source code are attached to this report.

2 | Section 1 - Transfer to the Parking Orbit

2.1. Introduction

The objective of this section is to design a transfer from a generic geocentric orbit, referred to as the initial orbit (i), to the parking orbit, referred to as the final orbit (f). The spacecraft remains in the parking orbit until the appropriate time to perform the departure burn required to inject it onto the Earth-departure hyperbolic trajectory discussed in Section 3.
Two alternative transfer strategies are considered and compared:

1. a transfer performed through a sequence of standard orbital maneuvers: an orbital-plane change, a change in the argument of periapsis, and an orbital-shape change;
2. a transfer achieved through a near-single-impulse maneuver, made possible by the near-intersection of the two orbits under consideration.

The adopted reference frame is the geocentric equatorial inertial frame.
The Earth's gravitational parameter is $\mu = 398\ 600.4\ km^3/s^2$.

2.2. Initial and Final Orbit data

For the initial orbit, the orbital parameters and the true anomaly at the initial time are known.

a_i $[km]$	e_i [-]	i_i $[deg]$	Ω_i $[deg]$	ω_i $[deg]$	θ_{start} $[deg]$
24.400×10^3	0.7283	6.00	80.21	178.02	85.20

The position and velocity vectors at the initial time are determined using the $par2car$ function.

$$[\boldsymbol{r_{start}}, \boldsymbol{v_{start}}] = par2car(a_i, e_i, i_i, \Omega_i, \omega_i\ , \theta_{start}, \mu)$$

$\boldsymbol{r_{start}}$ = (10.293, -3.070, -1.121)$\times 10^3$ km $\qquad$ $\boldsymbol{v_{start}}$ = (5.860, 4.782, -0.521) km/s

For the final orbit, the position and velocity vectors at the final time are known.

$\boldsymbol{r_{end}}$ = (-5.986, -7.931, 6.09)$\times 10^3$ km $\qquad$ $\boldsymbol{v_{end}}$ = (4.532, -5.96, -2.832) km/s

The inverse function, $car2par$, is used to determine the orbital parameters.

$$[a_f, e_f, i_f, \Omega_f, \omega_f, \theta_{end}] = car2par(\boldsymbol{r_{end}}, \boldsymbol{v_{end}}, \mu)$$

a_f $[km]$	e_f [-]	i_f $[deg]$	Ω_f $[deg]$	ω_f $[deg]$	θ_{end} $[deg]$
46.750×10^3	0.7507	39.65	100.67	125.00	0

The two orbits do not intersect [Figure 2.1], although they pass very close to each other at certain points. This property is exploited to define the near-single-impulse transfer.

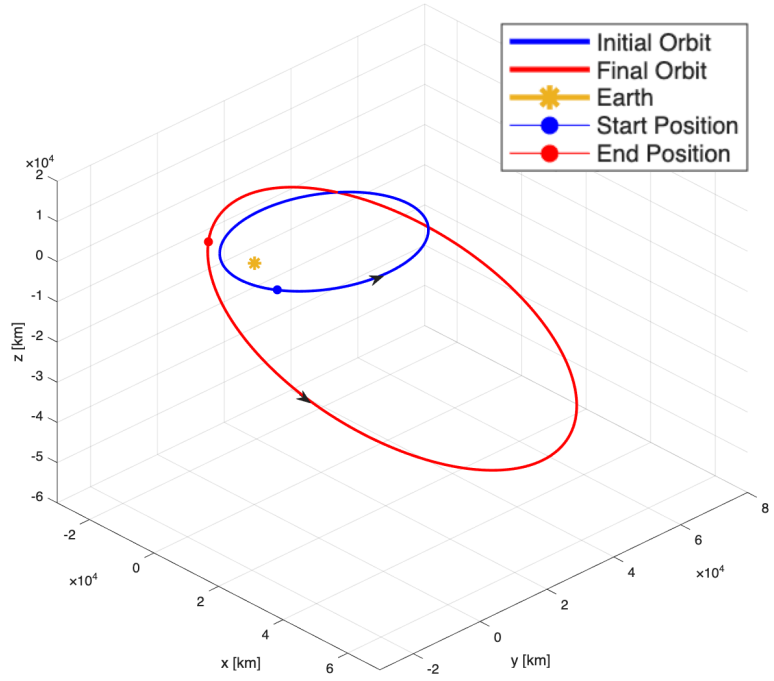


Figure 2.1

2.3. Transfer Using Standard Maneuvers

In this section, the transfer is performed through a standard sequence of three orbital maneuvers. Three intermediate transfer orbits are introduced to change the orbital plane, the argument of periapsis, and the orbital shape, respectively. Since this is a well-established procedure, only the results are presented, without a detailed discussion of the underlying method.

2.3.1. Orbital Plane Change

We now characterize the first transfer orbit [Figure 2.2]. It has the same shape as the initial orbit but lies in the plane of the final orbital. It is obtained using the *changeOrbitalPlane* function.

$$a_{T1} = a_i \qquad e_{T1} = e_i \qquad i_{T1} = i_f \qquad \Omega_{T1} = \Omega_f$$

$$[\Delta v_1,\, \omega_{T1},\, \theta_{M1}] = changeOrbitalPlane(a_i,\, e_i,\, i_i,\, \Omega_i,\, \omega_i,\, i_{T1},\, \Omega_{T1},\, \mu)$$

$$\Delta v_1 = 1.183\ km/s \qquad \omega_{T1} = 158.30^\circ \qquad \theta_{M1} = 205.43^\circ$$

2.3.2. Change in the Argument of Periapsis

We now characterize the second transfer orbit [Figure 2.3]. It has the same shape as the initial orbit, lies in the plane of the final orbit, and is apsidally aligned with it. The maneuver can be performed at either of two points with the same Δv cost. We therefore select the maneuver point closest to periapsis to minimize the transfer time. The maneuver parameters are computed using the *changePericenterArg* function.

$$a_{T2} = a_{T1} \qquad e_{T2} = e_{T1} \qquad i_{T2} = i_{T1} \qquad \Omega_{T2} = \Omega_{T1}$$

$$[\Delta v_2,\, \boldsymbol{\theta_{T1}},\, \boldsymbol{\theta_{T2}}] = changePericenterArg(a_{T1},\, e_{T1},\, \omega_{T1},\, \omega_f)$$

$$\Delta v_2 = 2.461\ km/s \qquad \theta_{T1,M2} = \boldsymbol{\theta_{T1}}(1) = 343.35^\circ \qquad \theta_{T2,M2} = \boldsymbol{\theta_{T2}}(1) = 16.65^\circ$$

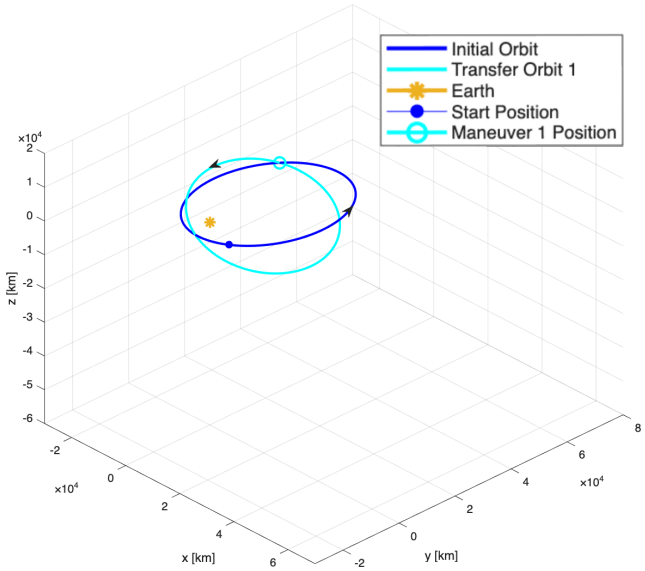


Figure 2.2

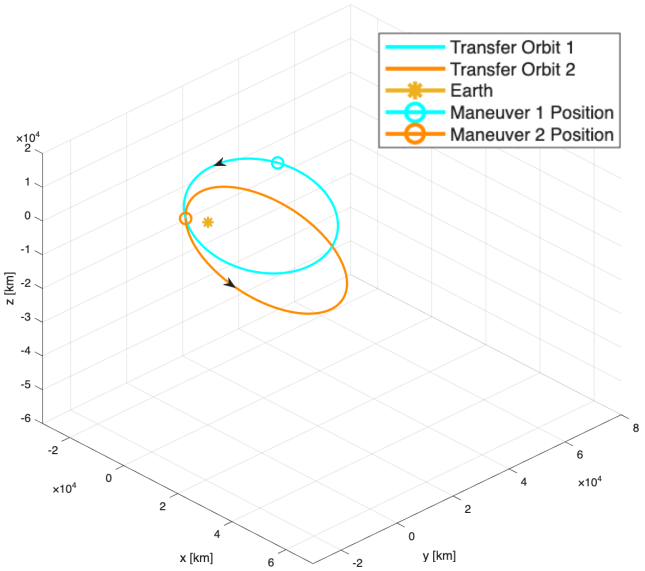


Figure 2.3

2.3.3. Orbital Shape Change

Since $\frac{a_f}{a_i} \approx 2$, we adopt a bitangent apoapsis-to-periapsis transfer. We therefore characterize the third transfer orbit [Figures 2.4 and 2.5]. Its apoapsis coincides with that of the initial orbit, while its periapsis coincides with that of the final orbit. The corresponding maneuver costs are computed using the $bitangentTransfer$ function.

$$r_{a,T3} = r_{a,T2} \qquad r_{p,T3} = r_{p,f} \qquad i_{T3} = i_{T2} \qquad \Omega_{T3} = \Omega_{T2} \qquad \omega_{T3} = \omega_{T2}$$

$$[\Delta v_3,\, \Delta v_4] = bitangentTransfer(a_{T2},\, e_{T2},\, a_f,\, e_f,\, \text{'ap'},\, \mu)$$

$$\Delta v_3 = 0.421\ km/s \qquad \Delta v_4 = 0.417\ km/s$$

2.3.4. Conclusion

The transfer time is computed as the sum of the times spent by the spacecraft on the three transfer orbits, using the $timeOfFlight$ function.

$$\Delta t_{T1} = timeOfFlight(a_{T1},\, e_{T1},\, \theta_{M1},\, \theta_{T1,M2},\, \mu) = 2.43\ h$$

$$\Delta t_{T2} = timeOfFlight(a_{T2},\, e_{T2},\, \theta_{T2,M2},\, \pi,\, \mu) = 5.21\ h$$

$$\Delta t_{T3} = timeOfFlight(a_{T3},\, e_{T3},\, \pi,\, 0,\, \mu) = 6.10\ h$$

$$\Delta t = \Delta t_{T1} + \Delta t_{T2} + \Delta t_{T3} = 13.75\ h$$

The total Δv cost of the transfer is:

$$\Delta v = \Delta v_1 + \Delta v_2 + \Delta v_3 + \Delta v_4 = 4.483\ km/s$$

The proposed maneuver sequence is not fully optimized. The following section presents an alternative strategy that reduces both the Δv cost and the transfer time.

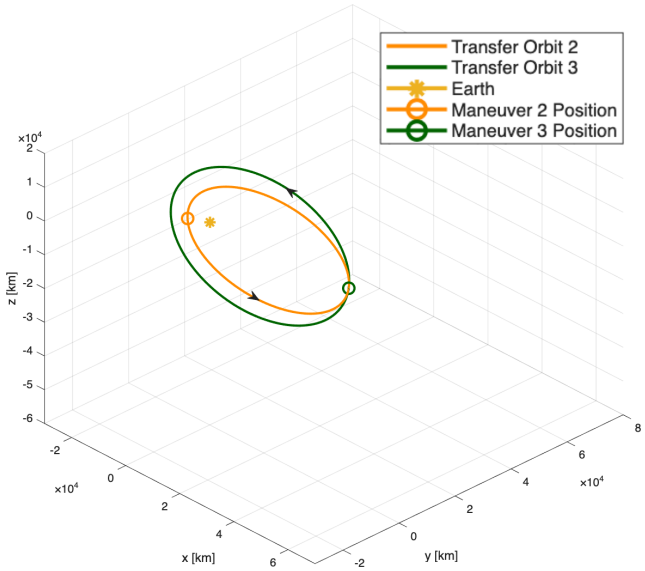


Figure 2.4

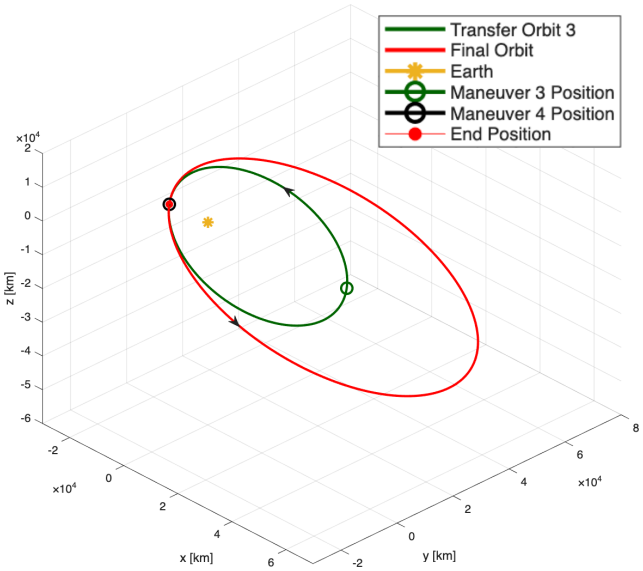


Figure 2.5

2.4. Near-Single-Impulse Maneuver

2.4.1. Introduction

The near-intersection of the initial and final orbits motivated an investigation into the feasibility of a near-single-impulse maneuver. The approach consists of applying a small correction to the initial orbit, thereby injecting the spacecraft into a very similar transfer orbit that intersects the final orbit. The adopted procedure is briefly outlined below.

We determine the point on the final orbit that lies in the plane of the initial orbit, denoted by $\boldsymbol{r_I}$. Two points satisfy this condition. Since the orbits nearly intersect, one of them lies very close to their point of closest approach and is therefore selected. We then impose the following constraints on the transfer orbit:

1. It passes through the point $\boldsymbol{r_I}$, where the main impulse is applied;
2. Its apoapsis radius is equal to that of the initial orbit: $r_{a,T} = r_{a,i}$;
3. It has the same orbital plane and orientation as the initial orbit: $i_T = i_i$, $\Omega_T = \Omega_i$, $\omega_T = \omega_i$.

These constraints allow the correction maneuver to be performed at the common apoapsis of the initial and transfer orbits. The $por2for$ function takes as input a position vector, an apoapsis radius, and the orbital orientation parameters, and returns the orbit's shape parameters and the true anomaly at the specified position. It is used to complete the characterization of the transfer orbit.

The cost of the correction maneuver at apoapsis, Δv_c, is computed as the absolute difference between the velocity magnitudes of the initial and transfer orbits at apoapsis. The cost of the main impulse, Δv_I, is computed as the norm of the difference between the velocity vectors on the transfer and final orbits at $\boldsymbol{r_I}$.

2.4.2. Determination of the Impulse Point

We determine the intersection of the final orbit with the plane of the initial orbit that lies near the point of closest approach between the two orbits. To this end, we define the unit angular momentum vector of the initial orbit, $\boldsymbol{h}$, and the position vector on the final orbit as a function of its true anomaly, $\boldsymbol{r_f}(\theta_f)$. We then introduce the scalar function f, defined as the dot product of these two vectors.The desired intersection point is obtained by finding a root of f.

$$\boldsymbol{h} = \frac{\boldsymbol{r}_{start} \times \boldsymbol{v}_{start}}{\|\boldsymbol{r}_{start} \times \boldsymbol{v}_{start}\|}$$

$$\boldsymbol{r_f}(\theta_f) = par2car(a_f,\, e_f,\, i_f,\, \Omega_f,\, \omega_f,\, \theta_f,\, \mu)$$

$$f(\theta_f) = \boldsymbol{h} \cdot \boldsymbol{r_f}(\theta_f)$$

An initial guess for the root associated with the solution near the point of closest approach between the two orbits is determined graphically. The root $\theta_{f,I}$ is then computed using the MATLAB function $fsolve$. The corresponding position vector $\boldsymbol{r}_I$ is then obtained using $par2car$.

$$\theta_{f,I} = 238.74^\circ$$

$$\boldsymbol{r_I} = par2car(a_f,\, e_f,\, i_f,\, \Omega_f,\, \omega_f\ ,\, \theta_{f,I},\, \mu)$$

$$\boldsymbol{r_I} = (-7.826,\, 32.467,\, 1.390) \times 10^3\ km$$

2.4.3. Characterization of the Transfer Orbit

We now characterize the transfer orbit [Figure 2.6]. To perform the correction maneuver at the common apoapsis of the initial and transfer orbits, we require the transfer orbit to pass through $\boldsymbol{r_I}$ and impose the following conditions:

$$r_{a,T} = r_{a,i} \qquad i_T = i_i \qquad \Omega_T = \Omega_i \qquad \omega_T = \omega_i$$

These constraints uniquely define the transfer orbit. Using the $por2for$ function, we determine its semi-major axis, eccentricity, and true anomaly at $\boldsymbol{r}_I$.

$$[a_T,\, e_T,\, \theta_{T,I}] = por2for(\boldsymbol{r_I},\, r_{a,T},\, i_T,\, \Omega_T,\, \omega_T)$$

$$a_T = 24.381 \times 10^3\ km \qquad e_T = 0.7297 \qquad \theta_{T,I} = 205.43^\circ$$

We obtain $\frac{a_i - a_T}{a_i} = 0.08\%$ and $\frac{e_i - e_T}{e_i} = -0.19\%$, confirming that the transfer orbit is virtually indistinguishable from the initial orbit. The cost of the correction maneuver is computed as the absolute difference between the velocity magnitudes of the initial and transfer orbits at apoapsis:

$$v_{a,i} = \sqrt{\mu \left(\frac{2}{r_{a,i}} - \frac{1}{a_i} \right)}$$

$$v_{a,T} = \sqrt{\mu \left(\frac{2}{r_{a,T}} - \frac{1}{a_T} \right)}$$

$$\Delta v_c = |v_{a,i} \text{ - } v_{a,T}| = 0.004\ km/s$$

As expected, the cost of the correction maneuver is negligible compared with that of the main maneuver, which is computed in the following subsection.

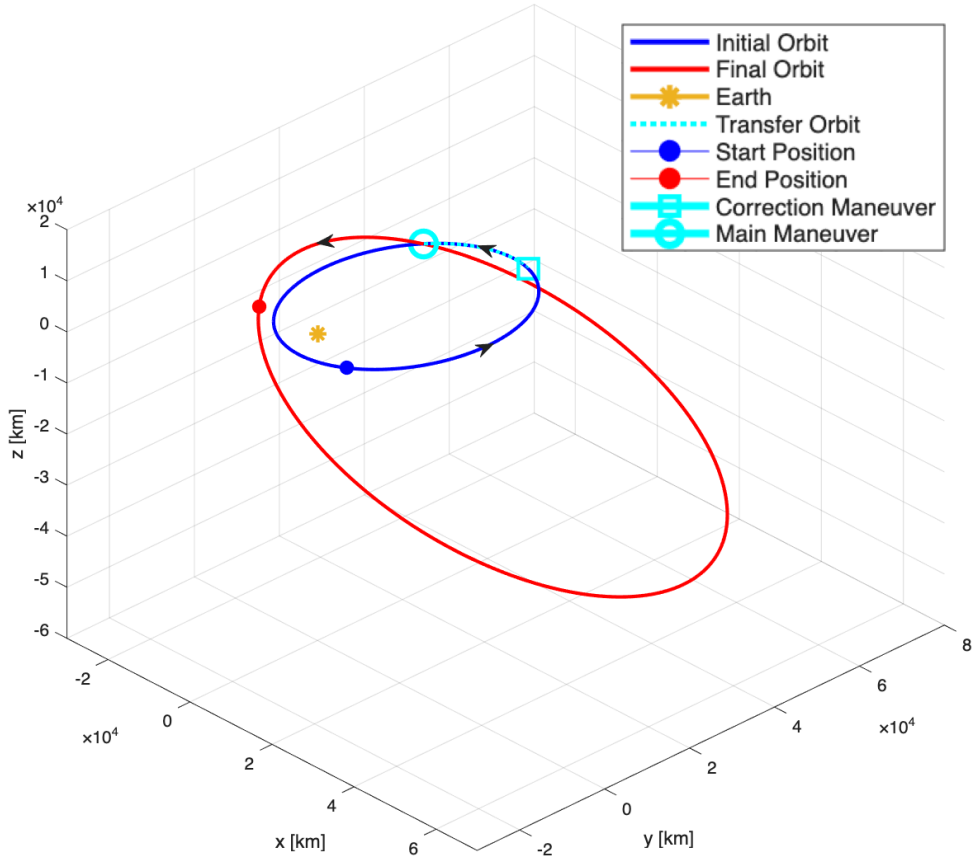


Figure 2.6

2.4.4. Conclusion

At $\boldsymbol{r}_I$, the true anomalies on both the transfer and final orbits are known. The cost of the main impulse is computed as the norm of the difference between the velocity vectors on the transfer and final orbits.

$$[\sim, \boldsymbol{v_{f,I}}] = par2car(a_f\ ,\ e_f\ ,\ i_f,\ \Omega_f,\ \omega_f,\ \theta_{f,I},\ \mu)$$

$$[\sim, \boldsymbol{v_{T,I}}] = par2car(a_T\ ,\ e_T\ ,\ i_T,\ \Omega_T,\ \omega_T,\ \theta_{T,I},\ \mu)$$

$$\Delta v_I = ||\boldsymbol{v_{f,I}} - \boldsymbol{v_{T,I}}|| = 1.817\ km/s$$

Since the correction maneuver has a negligible cost, the total Δv cost is approximately equal to that of the main impulse. The transfer time is computed as the time spent on the transfer orbit.

$$\Delta v = \Delta v_I + \Delta v_c = 1.821\ km/s \approx \Delta v_I$$

$$\Delta t = timeOfFlight(a_T,\ e_T,\ \pi,\ \theta_{T,I},\ \mu) = 2.79\ h$$

This transfer is more efficient than the standard-maneuver strategy in terms of both cost and transfer time. The resulting cost is expected to be close to the minimum required for this orbital transfer.
To verify this result, a direct transfer was designed using the algorithm developed for Section 2, allowing all three free variables, $\theta_{i,1}$, $\theta_{f,2}$, and ω_t, to vary (see the notation introduced in Section 2). The resulting minimum cost was approximately 1.81 km/s, with the maneuver again occurring near the point of closest approach between the two orbits.

3 Section 2 - Direct Transfer

3.1. Introduction

This section presents the design of a direct transfer orbit to the near-Earth asteroid 5797 Bivoj. The asteroid is classified as an Amor asteroid because its orbit approaches Earth's orbit from the outside without crossing it.

The objective is to design a mission that allows the spacecraft to reach and rendezvous with the asteroid. An essential preliminary step is therefore to obtain reliable ephemeris data describing the asteroid's actual position relative to Earth as a function of time.

According to data reported by Space Reference (https://www.spacereference.org/asteroid/5797-bivoj-1980-aa), the predicted closest approach of the asteroid to Earth will occur on 9 January 2092, at a distance of 10,476,346 km. As shown below, the minimum geometric distance between their orbits, computed from the prescribed orbital elements, is 10,538,261 km. The close agreement between these values suggests that, on that date, Earth and 5797 Bivoj will be located near the points of minimum separation on their respective orbits. We therefore adopt this configuration as the reference for the phasing analysis. The actual departure epoch is then determined by propagating the motion of the two bodies backward in time.

Although 9 January 2092 is far in the future, a mission of this type is most efficient when Earth and the asteroid are in a favorable configuration, characterized by a small separation and suitable orbital phasing. The purpose of this analysis is therefore to estimate the minimum Δv required to complete the mission and the corresponding transfer time. The procedure used to construct and optimize the transfer trajectory is outlined below.

The transfer geometry is parameterized by three free variables:

1. $\theta_{e,1}$: Earth's true anomaly at departure;
2. $\theta_{a,2}$: the asteroid's true anomaly at arrival;
3. ω_t: the argument of periapsis of the transfer orbit.

The subscripts 1 and 2 denote the departure and arrival conditions, respectively. By propagating the motion of the two bodies from the known reference configuration, each value of $\theta_{e,1}$ is associated with the corresponding value of $\theta_{a,1}$, namely the asteroid's true anomaly at the departure time. For a prescribed value of $\theta_{e,1}$ and the corresponding value of $\theta_{a,1}$, the remaining free variables are $\theta_{a,2}$ and ω_t. The transfer orbit is then determined by introducing the following two functions:

1. transfer cost: $\Delta v = \Delta v(\theta_{a,2}, \omega_t)$. The minimum of this function is sought;
2. delay in the spacecraft's passage through a given point on the asteroid's orbit relative to the asteroid's passage through the same point: $\Delta T = \Delta T(\theta_{a,2}, \omega_t)$. The root of this function is sought to ensure a rendezvous between the spacecraft and the asteroid.

By solving the system $\{\Delta T = 0;\ \min \Delta v\}$, we uniquely determine the transfer orbit that satisfies the rendezvous condition while minimizing the transfer cost.
The procedure is then repeated for different values of $\theta_{e,1}$ and the corresponding values of $\theta_{a,1}$ to identify the optimal departure configuration and further reduce the transfer cost.

The adopted reference frame is the heliocentric ecliptic frame.
The gravitational parameter used is that of the Sun: $\mu = 132\ 712\ 440\ 018\ \text{km}^3/\text{s}^2$.

3.2. Orbital Data

The orbital parameters of Earth's orbit (e) and the orbit of asteroid 5797 Bivoj (a) are reported below [Figure 3.1].

a_e $[km]$	e_e [-]	i_e $[deg]$	Ω_e $[deg]$	ω_e $[deg]$
149.46×10^6	0.0160	5.27×10^{-3}	159.55	301.62

a_a $[km]$	e_a [-]	i_a $[deg]$	Ω_a $[deg]$	ω_a $[deg]$
283.32×10^6	0.4443	4.19	298.91	168.65

3.2.1. Minimum-Distance Configuration

We determine the true anomalies of Earth and the asteroid corresponding to the minimum separation between their respective orbits. To this end, we define the distance between two points on the orbits as a function of their true anomalies and compute its minimum numerically.

$$\boldsymbol{r_e}\ (\theta_e) = par2car(a_e,\ e_e,\ i_e,\ \Omega_e,\ \omega_e\ ,\ \theta_e,\ \mu)$$

$$\boldsymbol{r_a}\ (\theta_a) = par2car(a_a,\ e_a,\ i_a,\ \Omega_a,\ \omega_a\ ,\ \theta_a,\ \mu)$$

$$\Delta r\ (\theta_e,\ \theta_a) = ||\boldsymbol{r_e}\ (\theta_e)\ \text{-}\ \boldsymbol{r_a}\ (\theta_a)||$$

An initial guess is obtained from the plot of this function and subsequently used to determine the minimum using the MATLAB function $fminsearch$.

$$\theta_{e,md} = 8.98^\circ \qquad \theta_{a,md} = 2.56^\circ \qquad \Delta r_{min} = 10,538,261\ km$$

Based on the considerations outlined in the introduction, we assume that Earth and 5797 Bivoj are in approximately this configuration on 9 January 2092..

3.3. Orbital Propagation

In this section, we define $\theta_{e,1}$ and $\theta_{a,1}$ as the true anomalies of Earth and the asteroid at the spacecraft departure time. Using orbital propagation and the known positions of the two bodies on 9 January 2092, we determine Bivoj's position as a function of Earth's position. We consider true anomalies smaller than those corresponding to the minimum-distance configuration, thereby placing the departure before the reference date. Δt_{adv} is the time by which the departure precedes the reference date.

$$\Delta t_{adv}(\theta_{e,1}) = timeOfFlight(a_e,\ e_e,\ \theta_{e,1},\ \theta_{e,md},\ \mu) \qquad \text{for } \theta_{e,1} \leq \theta_{e,md}$$

$$\theta_{a,1}(\theta_{e,1}) = orbitProp(a_a,\ e_a,\ \Delta t_{adv}(\theta_{e,1}),\ \theta_{a,md},\ \mu,\ 1).$$

In addition to the orbital and gravitational parameters, the function *orbitProp* takes a true anomaly and a time interval as inputs. The input value 1 indicates that $\theta_{a,1}$ is the true anomaly from which the asteroid reaches $\theta_{a,md}$ after the time interval Δt_{adv}. The input value 2 instead indicates forward propagation from the specified initial true anomaly.

Once the transfer time Δt has been determined, Earth's true anomaly at the time of the spacecraft's arrival at the asteroid can also be computed:

$$\theta_{e,2} = orbitProp(a_e,\, e_e,\, \Delta t,\, \theta_{e,1},\, \mu,\, 2).$$

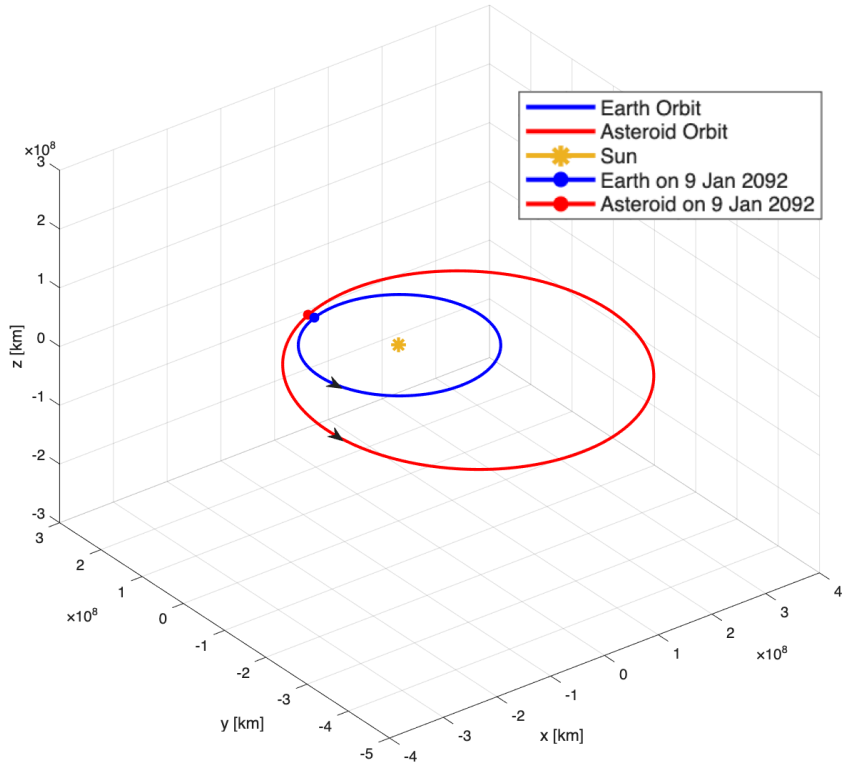


Figure 3.1

3.4. Characterization of the Transfer Orbit

Let $\theta_{e,1}$ (and the corresponding $\theta_{a,1}$) be prescribed. We now describe the algorithm used to determine the parameters of the minimum-cost transfer orbit that ensures rendezvous with the asteroid.

The free variables are $\theta_{a,2}$ and ω_t. We define the position and velocity vectors of Earth at departure ($\boldsymbol{r_1}$, $\boldsymbol{v_{e,1}}$) and of the asteroid at arrival ($\boldsymbol{r_2}$, $\boldsymbol{v_{a,2}}$).

$$[\boldsymbol{r_1},\, \boldsymbol{v_{e,1}}] = par2car(a_e,\, e_e,\, i_e,\, \Omega_e,\, \omega_e\ ,\, \theta_{e,1},\, \mu)$$

$$[\boldsymbol{r_2}(\theta_{a,2}),\, \boldsymbol{v_{a,2}}(\theta_{a,2})] = par2car(a_a,\, e_a,\, i_a,\, \Omega_a,\, \omega_a\ ,\, \theta_{a,2},\, \mu)$$

The two position vectors $\boldsymbol{r_1}$ and $\boldsymbol{r_2}$, together with the argument of periapsis ω_t, uniquely determine the transfer orbit. The function *direct* returns the remaining orbital parameters (a_t, e_t, i_t, Ω_t), as well as the true anomalies ($\theta_{t,1}$, $\theta_{t,2}$) corresponding to the positions $\boldsymbol{r_1}$ and $\boldsymbol{r_2}$. In this case, all these quantities are functions of $\theta_{a,2}$ and ω_t.

$$[a_t,\, e_t,\, i_t\ ,\, \Omega_t\ ,\, \theta_{t,1},\, \theta_{t,2}]\ (\theta_{a,2},\, \omega_t) = direct(\boldsymbol{r_1},\, \boldsymbol{r_2}(\theta_{a,2}),\, \omega_t)$$

3.4.1. Transfer Cost: $\Delta v(\theta_{a,2},\, \omega_t)$

We now define the cost function Δv to be minimized. The vectors $\boldsymbol{v_{t,1}}$ and $\boldsymbol{v_{t,2}}$ denote the spacecraft velocities on the transfer orbit at $\boldsymbol{r_1}$ and $\boldsymbol{r_2}$, respectively. Each contribution to the total cost is computed as the norm of the difference between the spacecraft velocity and that of the corresponding celestial body at the same position.

$$[\sim,\, \boldsymbol{v_{t,1}}(\theta_{a,2},\, \omega_t)] = par2car(a_t(\theta_{a,2},\, \omega_t)\ ,\ e_t(\theta_{a,2},\, \omega_t)\ ,\ i_t(\theta_{a,2},\, \omega_t),\ \Omega_t(\theta_{a,2},\, \omega_t),\ \omega_t,\ \theta_{t,1}(\theta_{a,2},\, \omega_t),\ \mu)$$

$$[\sim,\, \boldsymbol{v_{t,2}}(\theta_{a,2},\, \omega_t)] = par2car(a_t(\theta_{a,2},\, \omega_t)\ ,\ e_t(\theta_{a,2},\, \omega_t)\ ,\ i_t(\theta_{a,2},\, \omega_t),\ \Omega_t(\theta_{a,2},\, \omega_t),\ \omega_t,\ \theta_{t,2}(\theta_{a,2},\, \omega_t),\ \mu)$$

$$\Delta v_1(\theta_{a,2},\, \omega_t) = ||\boldsymbol{v_{t,1}}(\theta_{a,2},\, \omega_t) \text{ - } \boldsymbol{v_{e,1}}||$$

$$\Delta v_2(\theta_{a,2},\, \omega_t) = ||\boldsymbol{v_{t,2}}(\theta_{a,2},\, \omega_t) \text{ - } \boldsymbol{v_{a,2}}(\theta_{a,2})||$$

$$\Delta v(\theta_{a,2},\, \omega_t) = \Delta v_1(\theta_{a,2},\, \omega_t) + \Delta v_2(\theta_{a,2},\, \omega_t)$$

3.4.2. Time Delay: $\Delta T(\theta_{a,2},\, \omega_t)$

We now define the time-delay function ΔT, which is set to zero to enforce rendezvous. The quantity Δt_a denotes the time required by the asteroid to reach $\boldsymbol{r_2}$ from the position corresponding to $\theta_{a,1}$. The quantity Δt denotes the transfer time, namely the time required by the spacecraft to travel from $\boldsymbol{r_1}$ to $\boldsymbol{r_2}$. The function ΔT represents the delay of the spacecraft's passage through $\boldsymbol{r_2}$ relative to that of the asteroid; it is positive when the spacecraft arrives late and negative when it arrives early.

$$\Delta t_a(\theta_{a,2}) = timeOfFlight(a_a,\, e_a,\, \theta_{a,1},\, \theta_{a,2},\, \mu)$$

$$\Delta t(\theta_{a,2},\, \omega_t) = timeOfFlight(a_t(\theta_{a,2},\, \omega_t),\, e_t(\theta_{a,2},\, \omega_t),\, \theta_{t,1}(\theta_{a,2},\, \omega_t),\, \theta_{t,2}(\theta_{a,2},\, \omega_t),\, \mu)$$

$$\Delta T(\theta_{a,2},\, \omega_t) = \Delta t(\theta_{a,2},\, \omega_t) - \Delta t_a\ (\theta_{a,2})$$

3.4.3. Numerical Solution Procedure

The following procedure is used to solve the system:

$$\begin{cases} min \Delta v(\theta_{a,2}, \omega_t) \\ \Delta T(\theta_{a,2}, \omega_t) = 0 \end{cases}$$

1. The variable $\theta_{a,2}$ is assigned the value $\overline{\theta}_{a,2}$.

2. The admissible interval $(\omega_{min}, \omega_{max})$ of ω_t is determined such that the orbital parameters (a_t, e_t, i_t, Ω_t) define an elliptic orbit. We impose

$$0 < e_t(\overline{\theta}_{a,2},\, \omega_t) < 1 \qquad \text{for } \omega_t \in (\omega_{min},\, \omega_{max}).$$

3. The initial guess $\omega_{t,0}$ for the minimization is obtained from the plot of $\Delta v(\overline{\theta}_{a,2}, \omega_t)$, $\omega_t \in (\omega_{min}, \omega_{max})$.

4. A neighborhood Θ of $\overline{\theta}_{a,2}$ is defined over which the initial guess $\omega_{t,0}$ is assumed to remain valid; that is, it allows the following problem to be solved numerically for every $\theta_{a,2} \in \Theta$:

$$min \Delta v(\theta_{a,2},\, \omega_t)$$

5. We define the function $transferTime$, which, for each input $\tilde{\theta}_{a,2} \in \Theta$, solves

$$min \Delta v(\tilde{\theta}_{a,2},\, \omega_t) \qquad \text{for } \omega_t = \tilde{\omega}_t$$

using $fminsearch$ with the initial guess $\omega_{t,0}$ and then computes the orbital parameters

$$\tilde{a}_t = a_t(\tilde{\theta}_{a,2}\ ,\ \tilde{\omega}_t) \qquad \tilde{e}_t = e_t(\tilde{\theta}_{a,2}\ ,\ \tilde{\omega}_t) \qquad \tilde{\theta}_{t,1} = \theta_{t,1}(\tilde{\theta}_{a,2}\ ,\ \tilde{\omega}_t) \qquad \tilde{\theta}_{t,2} = \theta_{t,2}(\tilde{\theta}_{a,2}\ ,\ \tilde{\omega}_t)$$

It then returns the corresponding transfer time:

$$\Delta\tilde{t} = timeOfFlight(\tilde{a}_t,\, \tilde{e}_t,\, \tilde{\theta}_{t,1},\, \tilde{\theta}_{t,2},\, \mu)$$

Therefore, the function can be written as:

$$\Delta\tilde{t}\ (\tilde{\theta}_{a,2}) = transferTime(\theta_{e,1},\, \tilde{\theta}_{a,2},\, \omega_{t,0})$$

6. The delay function is then reduced to the single-variable function $\Delta\tilde{T}(\theta_{a,2}) = \Delta T(\theta_{a,2}, \omega_{t,\min \Delta v})$.

$$\Delta\tilde{T}(\theta_{a,2}) = \Delta\tilde{t}\ (\theta_{a,2}) - \Delta t_a\ (\theta_{a,2}) \qquad \forall\theta_{a,2}\ \in \Theta$$

The existence of a root within Θ is assessed graphically. If a root exists, an initial guess is obtained from the plot and the equation is solved using $fsolve$. If no root exists within Θ, the value of $\overline{\theta}_{a,2}$ must be changed. The root of the following equation therefore gives the required true anomaly:

$$\Delta\tilde{T}(\theta_{a,2}) = 0 \qquad \text{for } \theta_{a,2} = \theta_{a,2,s}$$

7. The solution procedure is completed by minimizing Δv.

$$min\Delta v(\theta_{a,2,s},\, \omega_t) \qquad \text{for } \omega_t = \omega_{t,s}$$

The solution is obtained using $fminsearch$ with the previously defined initial guess $\omega_{t,0}$.

An example calculation is provided in Appendix A.
Once the solution $(\theta_{a,2,s}, \omega_{t,s})$ has been found, the transfer can be fully characterized.

$$a_{t,s} = a_t(\theta_{a,2,s},\, \omega_{t,s}) \qquad e_{t,s} = e_t(\theta_{a,2,s},\, \omega_{t,s}) \qquad i_{t,s} = i_t(\theta_{a,2,s},\, \omega_{t,s}) \qquad \Omega_{t,s} = \Omega_t(\theta_{a,2,s},\, \omega_{t,s})$$

$$\theta_{t,1,s} = \theta_{t,1}(\theta_{a,2,s},\, \omega_{t,s}) \qquad \theta_{t,2,s} = \theta_{t,2}(\theta_{a,2,s},\, \omega_{t,s})$$

$$\Delta v_s = \Delta v(\theta_{a,2,s},\, \omega_{t,s}) \qquad \Delta t_s = \Delta t(\theta_{a,2,s},\, \omega_{t,s})$$

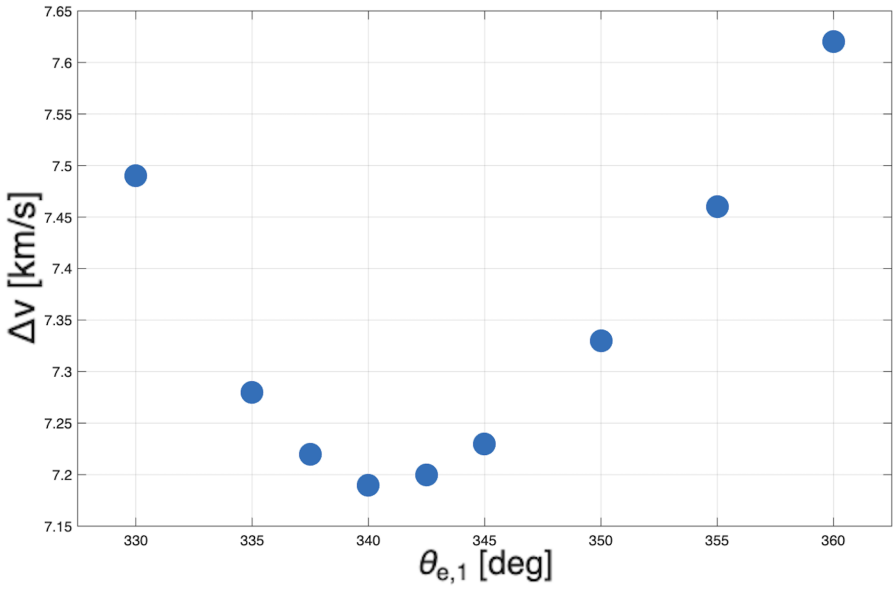


Figure 3.2

3.5. Conclusion

So far, the variable $\theta_{e,1}$ has been held fixed. The procedure described in Section 3.4 can be repeated for discrete values of $\theta_{e,1}$, yielding a different transfer orbit and corresponding transfer cost in each case. The transfer cost as a function of $\theta_{e,1}$ is shown in [Figure 3.2].

The curve reaches its minimum at $\theta_{e,1} = 340°$, which therefore represents the most favorable value of Earth's true anomaly at departure. The relevant parameters for this case and the corresponding orbital configuration are reported below [Figure 3.3].

a_t $[km]$	e_t [-]	i_t $[deg]$	Ω_t $[deg]$	ω_t $[deg]$	Δv $[km/s]$	Δt $[days]$
204.81×10^6	0.2813	2.37	261.05	180.07	7.193	68.24

$\theta_{e,1}$ $[deg]$	$\theta_{e,2}$ $[deg]$	$\theta_{a,1}$ $[deg]$	$\theta_{a,2}$ $[deg]$	$\theta_{t,1}$ $[deg]$	$\theta_{t,2}$ $[deg]$
340	49.35	332.05	43.51	0.05	69.89

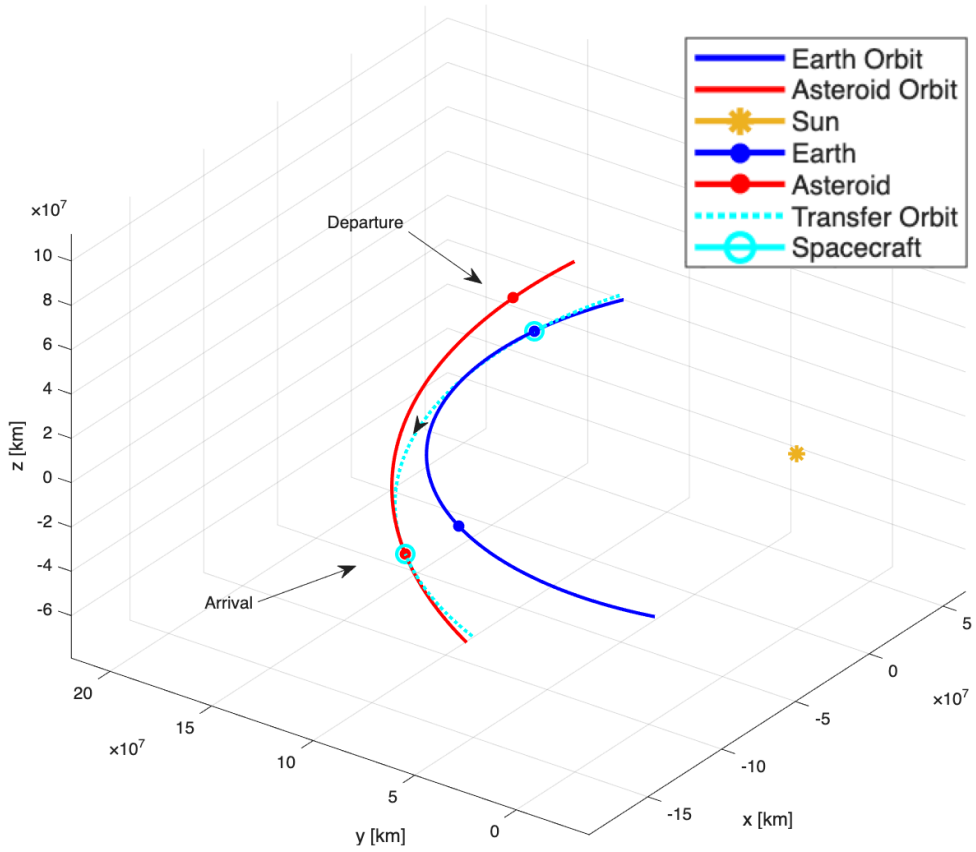


Figure 3.3

For $\theta_{e,1} = 340°$, we obtain $\Delta t_{adv} \approx 28$ days. The departure therefore occurs 28 days before 9 January 2092, corresponding to 12 December 2091. After a 68-day transfer, the spacecraft reaches the asteroid on 18 February 2092.

The proposed procedure is entirely general and is tied to neither the selected reference date nor the corresponding initial configuration. By changing this date, it is possible to identify earlier mission windows with more favorable Δv requirements.

4 | Section 3 - Hyperbolic Trajectories

4.1. Earth-Departure Hyperbola

4.1.1. Introduction

This section characterizes a hyperbolic departure trajectory connecting the parking orbit defined in Section 1 to the interplanetary transfer orbit obtained in Section 2. This trajectory enables the spacecraft to escape Earth's gravitational field in the direction and with the asymptotic velocity required for the subsequent mission phase.

The hyperbolic orbit is uniquely characterized by two parameters:

1. $\boldsymbol{v}_\infty$: the asymptotic velocity vector, whose value determined from the results of Section 2;
2. $\theta_{p,H}$: the true anomaly on the parking orbit at which the injection maneuver into the hyperbolic trajectory is performed. This parameter is the free variable used to minimize the maneuver cost.

Once these parameters have been specified, the hyperbolic orbit is characterized. This requires solving a nonlinear system of three equations in three unknowns.

The adopted reference frame is the geocentric equatorial inertial frame.
Earth's gravitational parameter is: $\mu = 398\ 600.4\ \text{km}^3/\text{s}^2$.

4.1.2. Asymptotic Velocity

The asymptotic velocity relative to Earth, expressed in the heliocentric ecliptic reference frame, is obtained by subtracting Earth's velocity from the spacecraft velocity on the transfer orbit. These velocity vectors were determined in Section 2.

$$\boldsymbol{v}_{\infty,helio} = \boldsymbol{v}_{t,1} - \boldsymbol{v}_{e,1} = (\text{-}3.635,\ 0.738,\ \text{-}1.406)\ km/s$$

Expressing this velocity in the geocentric equatorial reference frame requires a rotation through the obliquity angle $\epsilon = 23.45°$ about the x-axis, which points toward the vernal equinox. The corresponding rotation matrix is denoted by $\boldsymbol{R}_{\boldsymbol{x}}(\epsilon)$.

$$\boldsymbol{v}_\infty = \boldsymbol{R}_{\boldsymbol{x}}(\epsilon)\ \boldsymbol{v}_{\infty,elio} = (\text{-}3.635,\ 1.237,\ \text{-}0.996)\ km/s$$

The asymptotic escape direction and velocity magnitude can now be determined.

$$\boldsymbol{r}_\infty = \frac{\boldsymbol{v}_\infty}{\|\boldsymbol{v}_\infty\|} = (\text{-}0.916,\ 0.312,\ \text{-}0.251)\ [\text{-}]$$

$$v_\infty = ||\boldsymbol{v}_\infty|| = 3.967\ km/s$$

4.1.3. Parking Orbit and Maneuver Point

The orbital parameters of the parking orbit defined in Section 1 are reported below.

a_p $[km]$	e_p [-]	i_p $[deg]$	Ω_p $[deg]$	ω_p $[deg]$
46.750×10^3	0.7507	39.65	100.67	125.00

We prescribe the true anomaly $\theta_{p,H}$ corresponding to the maneuver point H on the parking orbit:

$$\theta_{p,H} = 177.5°$$

As shown in Section 4.1.5, varying this parameter allows the maneuver cost to be minimized.
The position vector corresponding to this true anomaly is computed using the $par2car$ function.

$$\boldsymbol{r_H} = par2car(a_p,\, e_p,\, i_p,\, \Omega_p,\, \omega_p,\, \theta_{p,H},\, \mu)$$
$$\boldsymbol{r_H} = (43.960,\, 52.905,\, -43.921)\times 10^3\ km$$

4.1.4. Characterization of the Hyperbolic Trajectory

The semi-major axis can be computed directly from the asymptotic velocity.

$$a_h = -\frac{\mu}{v_\infty^2} = -25.330 \times 10^3\ km$$

We also define the angle α between $\boldsymbol{r_H}$ and $\boldsymbol{r_\infty}$.

$$\alpha = arccos(\frac{\boldsymbol{r_H}}{\|\boldsymbol{r_H}\|} \cdot \boldsymbol{r_\infty}) = 99.00°$$

We now formulate a nonlinear system of three equations in three unknowns. Solving this system yields the eccentricity e_h, the true anomaly at the maneuver point $\theta_{h,H}$, and the asymptotic true anomaly θ_∞ of the hyperbolic orbit.

$$\begin{cases} f_1(e_h, \theta_{h,H}, \theta_\infty) = \frac{r_H}{a_h}(1 + e_h \cos(\theta_{h,H})) + e_h^2 - 1 \ = 0 \\ f_2(e_h, \theta_{h,H}, \theta_\infty) = \cos(\theta_\infty) + \frac{1}{e_h} \ = 0 \\ f_3(e_h, \theta_{h,H}, \theta_\infty) = \theta_\infty - \alpha - \theta_{h,H} \ = 0 \end{cases}$$

$$\boldsymbol{F}\ (e_h,\, \theta_{h,H},\, \theta_\infty) = \{\ f_1\ (e_h,\, \theta_{h,H},\, \theta_\infty),\ f_2\ (e_h,\, \theta_{h,H},\, \theta_\infty),\ f_3\ (e_h,\, \theta_{h,H},\, \theta_\infty)\ \} = \boldsymbol{0}$$

where $r_H = ||\boldsymbol{r_H}||$. We then determine a suitable initial guess for solving the system using $fsolve$.

- We assign an initial guess for the eccentricity: $e_{h,0} = 4$.
- We define the two-variable function $g\ (\theta_{h,H},\, \theta_\infty) = ||\boldsymbol{F}\ (e_{h,0},\, \theta_{h,H},\, \theta_\infty)||$.
- The pair $(\theta_{h,H}, \theta_\infty)$ that minimizes g provides a suitable initial guess. An approximate minimizer is sufficient: $\theta_{h,H,0} = 0.5\ rad$ and $\theta_{\infty,0} = 2.0\ rad$.

The system can now be solved using $fsolve$, yielding:

$$e_h = 4.2130 \qquad \theta_{h,H} = 4.73° \qquad \theta_\infty = 103.73°$$

The inclination i_h, the right ascension of the ascending node Ω_h, and the argument of periapsis ω_h remain to be determined. From the two known points on the hyperbolic orbit, $\boldsymbol{r_\infty}$ and $\boldsymbol{r_H}$, we compute

the unit angular momentum vector $\boldsymbol{h}$, from which i_h is obtained directly. We then determine the line of nodes $\boldsymbol{N}$ and Ω_h:

$$\boldsymbol{h} = \frac{\boldsymbol{r_H}}{||\boldsymbol{r_H}||} \times \boldsymbol{r_\infty} \qquad i_h = arccos(\boldsymbol{h} \cdot \boldsymbol{k}) = 39.51°$$

$$\boldsymbol{N} = \boldsymbol{k} \times \boldsymbol{h} \qquad \Omega_h = arccos(\boldsymbol{N} \cdot \boldsymbol{i}) = 179.54°$$

We complete the characterization of the hyperbolic orbit by computing ω_h and the other parameters of interest.

$$cos(\beta) = \boldsymbol{N} \cdot \frac{\boldsymbol{r_H}}{||\boldsymbol{r_H}||} \qquad sin(\beta) = (\boldsymbol{N} \times \frac{\boldsymbol{r_H}}{||\boldsymbol{r_H}||}) \cdot \boldsymbol{h} \qquad \omega_h = \beta \text{ - } \theta_{h,H} = 233.03°$$

$$\delta = 27.46° \qquad \Delta = 103.667 \times 10^3 \ km \qquad r_p = 81.387 \times 10^3 \ km$$

The maneuver cost is computed as the norm of the difference between the velocity vectors on the parking orbit and the departure hyperbola at the maneuver point:

$$[\sim, \boldsymbol{v_{p,H}}] = par2car(a_p, e_p, i_p, \Omega_p, \omega_p, \theta_{p,H}, \mu)$$

$$[\sim, \boldsymbol{v_{h,H}}] = par2car(a_h, e_h, i_h, \Omega_h, \omega_h, \theta_{h,H}, \mu)$$

$$\Delta v = ||\boldsymbol{v_{h,H}} \text{ - } \boldsymbol{v_{p,H}}|| = 4.377 \ km/s$$

4.1.5. Conclusion

So far, the parameter $\theta_{p,H}$ has been held fixed. We repeat the procedure described in Sections 4.1.3 and 4.1.4 for several discrete values of $\theta_{p,H}$, obtaining a different departure hyperbola and corresponding maneuver cost for each value. The variation of Δv with $\theta_{p,H}$ is shown in [Figure 4.1].

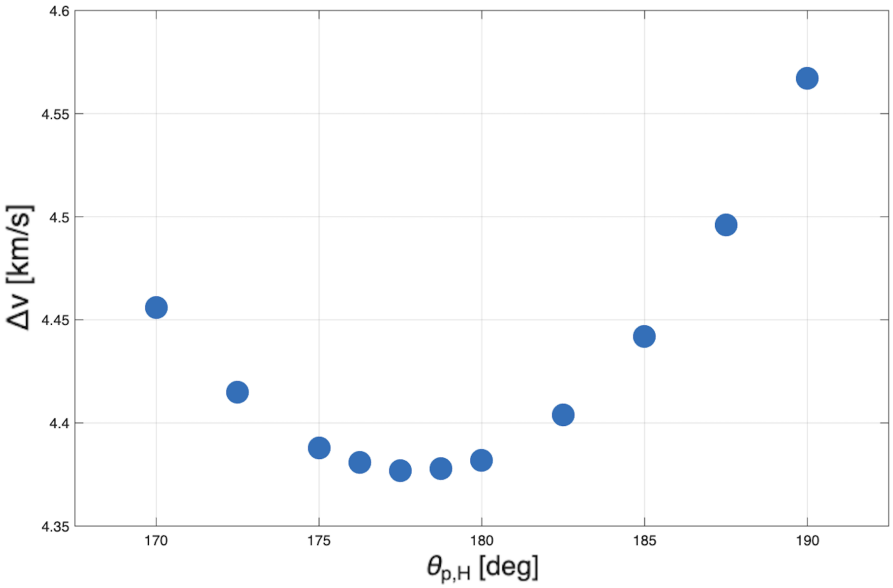


Figure 4.1

The curve reaches its minimum at $\theta_{p,H} = 177.5°$, which therefore represents the most favorable true anomaly for the maneuver. The parking orbit and the optimized departure hyperbola are illustrated in [Figure 4.2].

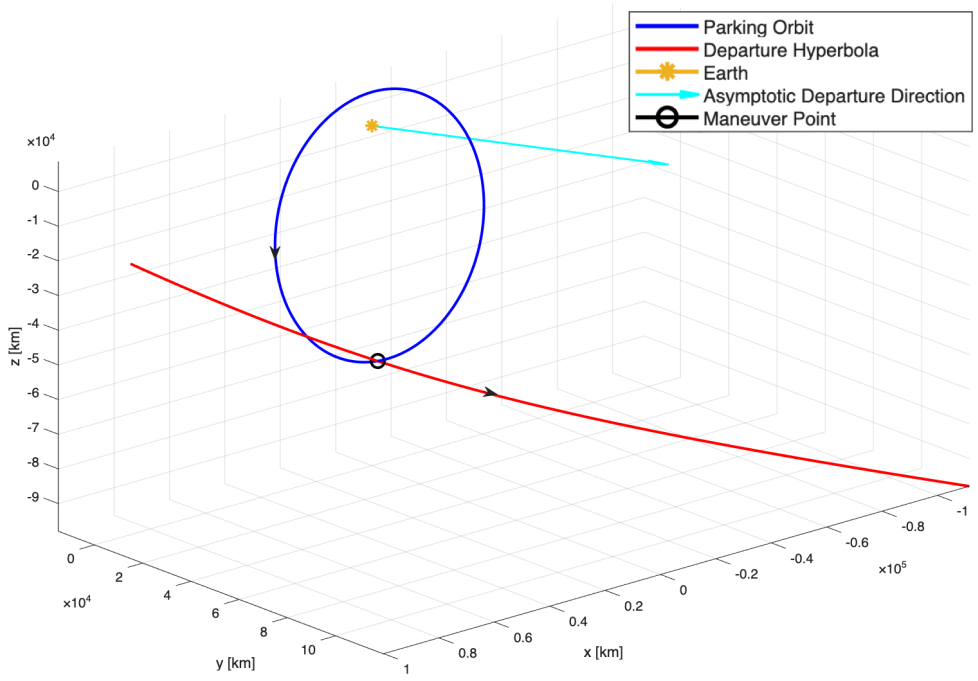


Figure 4.2

4.2. Asteroid-Arrival Hyperbola

4.2.1. Introduction

This section characterizes the asteroid-arrival hyperbola and the associated capture orbit. This analysis is required to model the transition of the spacecraft from the interplanetary trajectory to a bound orbit around the asteroid. The objective is to insert the spacecraft into a circular asteroid-centered orbit, while the design of a possible landing operation is left for future work. Unlike in the previous section, the spacecraft's actual arrival direction is not explicitly considered, since the capture orbit is assumed to be coplanar with the arrival hyperbola.

Within this simplified model, the arrival hyperbola is characterized by specifying the following two parameters:

- hyperbolic excess velocity v_∞, whose value is fixed and determined from the results of Section 2;
- periapsis radius r_p, which is the only free parameter to be selected.

The capture orbit is defined as a circular orbit of radius r_p that is coplanar with the arrival hyperbola. The insertion maneuver is therefore performed at the periapsis of the hyperbola.

The periapsis radius r_p is constrained by the geometry of the asteroid: it must be sufficiently greater than the body's mean radius of 0.2 km, while remaining small enough to prevent the spacecraft from orbiting excessively far from the surface. Based on comparisons with previous asteroid missions, the following reference interval is assumed: $1\ km < r_p < 10\ km$.

The asteroid's gravitational parameter can be computed from its mass.

$$M = 6.702 \times 10^{10}\ kg \qquad \mu = GM = 4.4732 \times 10^{-9}\ km^3/s^2$$

Since the gravitational parameter is extremely small, the problem approaches a limiting case in which the asteroid's gravitational effects are negligible relative to the other kinematic and dynamic effects

involved. It is therefore convenient to treat μ as a free variable and analyze the behavior of the relevant parameters in the limit $\mu \to 0$. This approach highlights the asymptotic properties of the resulting relationships and simplifies their physical interpretation.

The quantities characterizing the capture maneuver are functions of μ and r_p. In particular, we show that they depend on these variables through the ratio $\frac{\mu}{r_p}$. Since r_p remains finite within the operational interval considered, while μ is extremely small, we can consider

$$\frac{\mu}{r_p} \to 0 \qquad \text{for } \mu \to 0.$$

In particular, we show that the cost of the capture-orbit insertion maneuver, Δv, also depends on the ratio $\frac{\mu}{r_p}$ and becomes asymptotically independent of r_p. Consequently, the periapsis radius can be selected on the basis of other operational requirements.

4.2.2. Hyperbolic Excess Velocity

The hyperbolic excess velocity relative to the asteroid, expressed in the heliocentric ecliptic reference frame, is obtained by subtracting the spacecraft velocity on the transfer orbit from the asteroid's velocity. Since only its magnitude is required, the order of the subtraction is irrelevant. The two velocity vectors were determined in Section 2.

$$\boldsymbol{v_{\infty,helio}} = \boldsymbol{v_{a,2}} - \boldsymbol{v_{t,2}} \qquad v_\infty = ||\boldsymbol{v_{\infty,elio||}} = 3.226 \ km/s$$

4.2.3. Asymptotic Parameters

Given the hyperbolic excess velocity, the semi-major axis of the arrival hyperbola can be determined directly.

$$a_h(\mu) = -\frac{\mu}{v_\infty^2} \to 0 \qquad \text{for } \mu \to 0$$

We now express the eccentricity of the hyperbola as a function of the ratio $\frac{\mu}{r_p}$, from which the asymptotic behavior of the other hyperbolic parameters can be derived.

$$e_h(\tfrac{\mu}{r_p}) = 1 - \frac{r_p}{a_h(\mu)} = 1 + \frac{v_\infty^2}{\mu/r_p} \to \infty \qquad \text{for } \frac{\mu}{r_p} \to 0$$

$$\theta_\infty \to 90° \qquad \delta \to 0 \qquad \Delta \to r_p$$

As expected, the asteroid's mass is too small to significantly alter the motion of a nearby spacecraft traveling at speeds typical of heliocentric orbits.
The maneuver cost Δv can now be defined as the difference between the periapsis speed on the arrival hyperbola, $v_{p,h}$, and the speed on the circular capture orbit, v_c.

$$v_{p,h}(\tfrac{\mu}{r_p}) = \sqrt{v_\infty^2 + 2\frac{\mu}{r_p}} \qquad v_c(\tfrac{\mu}{r_p}) = \sqrt{\frac{\mu}{r_p}}$$

$$\Delta v(\tfrac{\mu}{r_p}) = v_{p,h}(\tfrac{\mu}{r_p}) - v_c(\tfrac{\mu}{r_p}) = \sqrt{v_\infty^2 + 2\frac{\mu}{r_p}} - \sqrt{\frac{\mu}{r_p}} \quad \to \quad v_\infty \qquad \text{for } \frac{\mu}{r_p} \to 0$$

The maneuver cost approaches the asymptotic speed and, over the operational range considered for the periapsis radius, is essentially independent of r_p. This result is consistent with the asteroid's extremely small mass, which has a negligible effect on the spacecraft's velocity. Consequently, the maneuver essentially consists of reducing the spacecraft's velocity relative to the asteroid to an almost negligible value, so that $\Delta v \approx v_\infty = 3.226 \ km/s$.

4.2.4. Periapsis Radius

The periapsis radius cannot be determined from the previous result and may therefore be selected within the reference interval according to additional operational requirements. In particular, the periapsis radius, which is also the radius of the circular capture orbit, should be sufficiently large to maintain orbital stability despite perturbations caused by the asteroid's irregular shape. At the same time, it should remain small enough not to compromise a possible subsequent landing operation.

The orbital configuration obtained for $r_p = 2.5\ km$ is shown in [Figure 4.3]. The hyperbolic trajectory is essentially indistinguishable from a straight line, as expected given the negligible deflection produced by the asteroid's gravitational field.

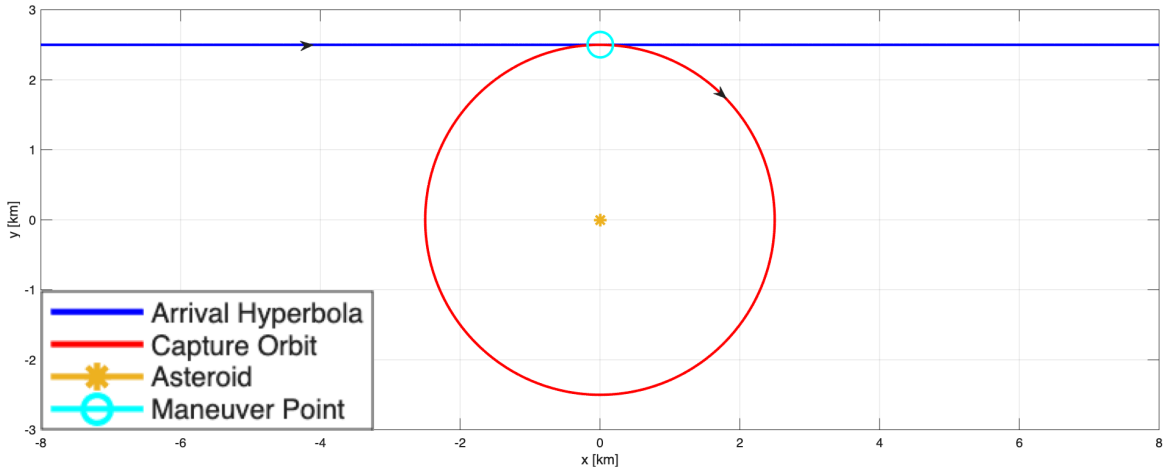


Figure 4.3

From an energetic perspective, arrival at the asteroid does not constitute gravitational capture but rather a deep-space rendezvous maneuver. The spacecraft is not naturally captured by the asteroid's weak gravitational attraction; instead, it must actively reduce its relative velocity to a near-zero value before entering orbit around the asteroid.

4.3. Conclusion

The total mission cost can now be computed using the patched-conic approximation.

$$\Delta v_{patched-conic} = \Delta v_{Earth\ departure} + \Delta v_{asteroid\ arrival} = (4.377 + 3.226)\ km/s = 7.603\ km/s$$

Within the assumptions of this preliminary analysis, the resulting Δv suggests that the proposed mission is energetically feasible and is consistent with the order of magnitude expected for interplanetary missions to near-Earth asteroids. However, 5797 Bivoj cannot be regarded as one of the most energetically accessible targets, since it requires a larger Δv budget than more favorable asteroids.

The cost obtained using the patched-conic approximation differs by only $0.410\ km/s$ from that of the direct heliocentric transfer. This suggests that the dominant contribution to the mission cost is associated with the geometry of the interplanetary transfer, while the additional cost introduced by the patched-conic modeling of the Earth-departure phase remains limited.

5 | Conclusion

This study assessed the preliminary feasibility and main operational characteristics of an interplanetary mission to the near-Earth asteroid 5797 Bivoj. The patched-conic analysis yielded a total Δv requirement of 7.603 km/s. Comparison with previous asteroid missions suggests that this requirement is compatible with current propulsion capabilities, supporting the technical feasibility of the proposed mission at a preliminary-design level.

5.0.1. Propulsion System

For a mission to a near-Earth asteroid with a Δv requirement of approximately 7.5 km/s and a transfer time of a few months, propulsion-system selection requires a trade-off among mission duration, propellant efficiency, spacecraft mass, power requirements, and overall cost.

The most established propulsion options for this type of mission are chemical and electric propulsion. Electric propulsion, employed in missions such as Hayabusa and Dawn, provides high specific impulse and therefore uses propellant very efficiently. However, its low thrust generally makes it more suitable for mission profiles involving longer transfer times and extended thrusting periods.

For a relatively short mission based on the impulsive-maneuver approximation adopted in this study, the ability to generate high thrust becomes a primary requirement. Chemical propulsion, used in missions such as NEAR Shoemaker and DART, is therefore better suited to the proposed mission profile, as it can produce the rapid velocity changes required by the planned maneuvers.

Chemical propulsion is also a mature and reliable technology with comparatively limited power requirements. Nevertheless, its lower specific impulse results in greater propellant consumption than electric propulsion. The final selection would therefore require detailed spacecraft mass and power budgets and could also consider a hybrid architecture combining high-thrust chemical maneuvers with more efficient electric propulsion during the interplanetary cruise.

5.0.2. Future Work

The simplifying assumptions adopted in this study imply that the resulting mission parameters should be regarded as preliminary estimates rather than exact predictions for a real mission. Future work could improve the fidelity of the analysis by incorporating more advanced models and additional physical effects, including:

- *N-body dynamics*: gravitational perturbations from other relevant celestial bodies are included. For an interplanetary mission, these may include the planets of the Solar System and any other bodies capable of significantly perturbing the trajectory.
- *Thermal constraints*: the spacecraft trajectory must maintain suitable distances from the Sun according to the technologies employed, including photovoltaic panels, thermal-control coatings, and the allowable operating-temperature range.

- *Continuous thrust*: the impulsive-maneuver approximation is replaced by a model in which the propulsion system produces finite acceleration over nonzero time intervals.
- *Relativistic corrections*: corrections to classical Newtonian mechanics may be included when required by the navigation accuracy, mission duration, or proximity to massive bodies.
- *Non-gravitational perturbations*: effects such as solar-radiation pressure on the spacecraft may be included. For long-term propagation of the asteroid's orbit, thermal effects such as the Yarkovsky effect may also be relevant.

5.0.3. Mission Rationale

The motivations for a mission to 5797 Bivoj can be divided into three categories:

- *Scientific research: Solar System archaeology.* Asteroids preserve valuable information about the formation of the early Solar System. Subsurface-penetrating radar, imaging systems, and spectrometers could be used to investigate Bivoj's internal structure and composition.
- *Technological and economic interest: resource prospecting.* A mission to Bivoj could assess potentially useful materials and test technologies for asteroid proximity operations and sample collection. However, the economic exploitation of such resources would require dedicated compositional measurements and major technological developments.
- *Planetary defense.* The growing number of known near-Earth asteroids has increased the importance of detection, characterization, and risk-mitigation technologies. Events such as the Chicxulub impact and the 2013 Chelyabinsk airburst demonstrate the potential consequences of asteroid impacts. Missions to near-Earth asteroids can also validate navigation, rendezvous, and characterization techniques relevant to planetary defense.

Asteroid exploration is a strategic field for the world's major space agencies, with several relevant missions currently in progress or under development:

- NASA is conducting the Lucy mission to study Jupiter's Trojan asteroids, while the Psyche spacecraft is traveling toward the metal-rich asteroid 16 Psyche. NASA is also developing NEO Surveyor, a space telescope designed to detect and characterize potentially hazardous near-Earth objects.
- The European Space Agency's Hera spacecraft is approaching the Didymos binary-asteroid system. Its objective is to study Dimorphos and assess the consequences of the impact produced by NASA's DART mission, thereby improving our understanding of asteroid-deflection techniques.
- China's Tianwen-2 spacecraft reached the near-Earth asteroid 2016 HO3 in July 2026 and began close-proximity observations in preparation for sample collection. The mission is subsequently expected to continue toward the main-belt comet 311P/PANSTARRS.

Asteroid exploration is likely to play an increasingly important role in future space activities. Its value lies not only in the scientific knowledge it can provide, but also in the development of technologies for autonomous navigation, planetary defense, resource assessment, and deep-space operations. From this perspective, asteroid missions serve a broader strategic purpose: they enable the exploration of new celestial bodies while building the expertise and infrastructure required for a safer and more sustained human presence in space.

A | Appendix A: Numerical Example

We present an example application of the procedure described in Section 3.4.3.
Let $\theta_{e,1} = 340°$, corresponding to $\theta_{a,1} = 332.05°$.

1. We set $\overline{\theta}_{a,2} = \frac{\pi}{4}$.

2. From the plot of $e_t(\overline{\theta}_{a,2}, \omega_t)$ over $\omega_t \in [0, 2\pi]$ [Figure A.1], we obtain $\omega_{min} \approx 0.7$ and $\omega_{max} \approx 3.5$.

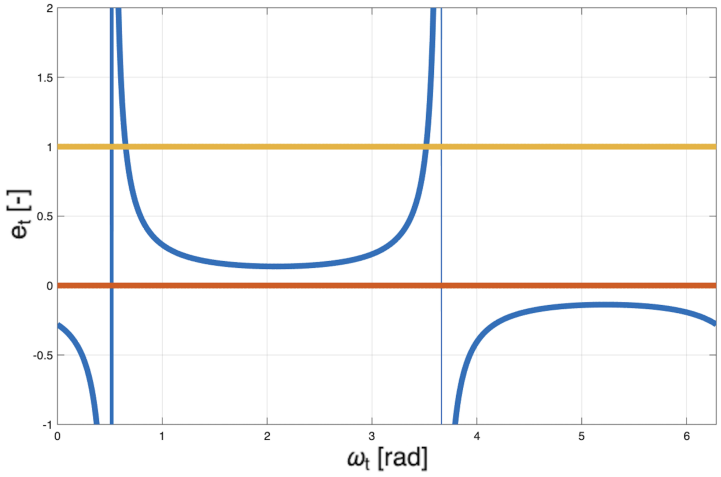


Figure A.1

3. From the plot of $\Delta v(\overline{\theta}_{a,2}, \omega_t)$ [Figure A.2] for $\omega_t \in (\omega_{min}, \omega_{max})$, we obtain the $\omega_{t,0} = 3.2$.

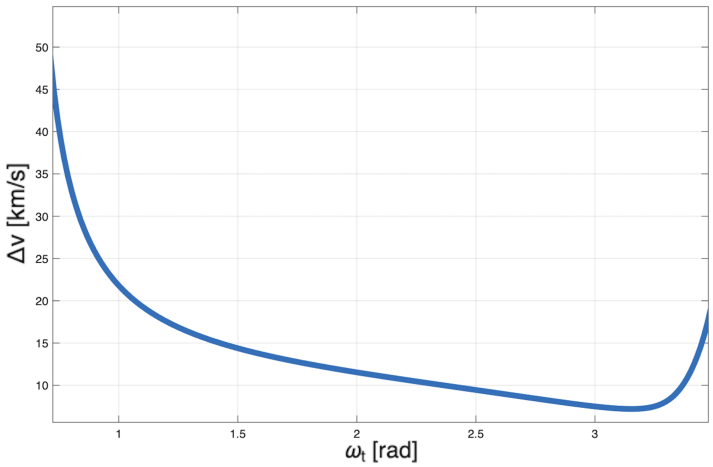


Figure A.2

4. We define $\Delta\theta = \frac{\pi}{12}\ rad$ and $\Theta = [\overline{\theta}_{a,2} - \Delta\theta\ ,\ \overline{\theta}_{a,2} + \Delta\theta]$.

5. We define $\Delta\tilde{t}_t(\theta_{a,2}) = transferTime(\theta_{e,1},\ \theta_{a,2},\ \omega_{t,0})$ for $\theta_{a,2} \in \Theta$.

6. We define $\Delta\tilde{T}(\theta_{a,2}) = \Delta\tilde{t}_t(\theta_{a,2}) - \Delta t_a(\theta_{a,2})$ for $\theta_{a,2} \in \Theta$, and plot the resulting function [Figure A.3].

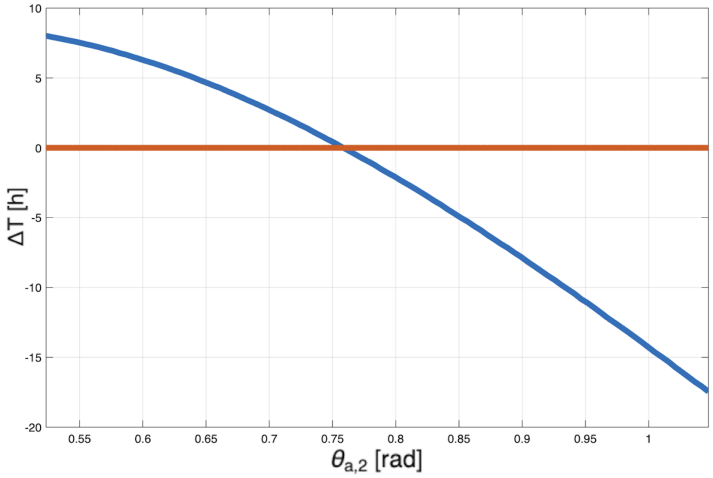


Figure A.3

A root exists within Θ.
From a numerical perspective, it is more efficient to minimize $|\Delta\tilde{T}(\theta_{a,2})|$. This is performed using $fminsearch$ with the initial guess $\theta_{a,2,0} = 0.75$, yielding $\theta_{a,2,s} = 0.7589$.

7. We then minimize the function $\Delta v(\theta_{a,2,s},\ \omega_t)$ using $fminsearch$ with the previously defined initial guess $\omega_{t,0}$. This yields $\omega_{t,s} = 3.1427$.

Therefore, $(\theta_{a,2,s},\ \omega_{t,s}) = (0.7589,\ 3.1427)$ is the solution to the problem.

B | Appendix B: Summary of Results

B.1. Section 1: Transfer to the Parkink Orbit

B.1.1. Standard Maneuver Sequence

Orbit	a $[km]$	e [-]	i $[deg]$	Ω $[deg]$	ω $[deg]$	Δt $[h]$
initial (i)	24.400×10^3	0.7283	6.00	80.21	178.02	-
transfer 1 $(T1)$	24.400×10^3	0.7283	39.65	100.67	158.30	2.43
transfer 2 $(T2)$	24.400×10^3	0.7283	39.65	100.67	125.00	5.21
transfer 3 $(T3)$	26.913×10^3	0.5669	39.65	100.67	125.00	6.10
final (f)	46.750×10^3	0.7507	39.65	100.67	125.00	-

Maneuver	previous o.	θ_{prev} $[deg]$	subsequent o.	θ_{sub} $[deg]$	Δv $[km/s]$
M1 - Plane-Change	i	85.20	$T1$	205.43	1.183
M2 - Argument-of-Periapsis change	$T1$	343.35	$T2$	16.65	2.461
M3 - Bitangent Tranfer (Apoapsis)	$T2$	180.00	$T3$	180.00	0.421
M4 - Bitangent Tranfer (Periapsis)	$T3$	0	f	0	0.417

$$\Delta v_{tot} = 4.483\ km/s \qquad \Delta t_{tot} = 13.75\ h$$

B.1.2. Near-Single-Impulse Maneuver

Orbit	a $[km]$	e [-]	i $[deg]$	Ω $[deg]$	ω $[deg]$	Δt $[h]$
initial (i)	24.400×10^3	0.7283	6.00	80.21	178.02	-
transfer (T)	24.381×10^3	0.7297	6.00	80.21	178.02	2.79
final (f)	46.750×10^3	0.7507	39.65	100.67	125.00	-

Maneuver	previous o.	θ_{prev} $[deg]$	subsequent o.	θ_{sub} $[deg]$	Δv $[km/s]$
c - Correction	i	180.00	T	180.00	0.004
I - Main Impulse	T	205.43	f	238.74	1.817

$$\Delta v_{tot} = 1.821\ km/s \qquad \Delta t_{tot} = 2.79\ h$$

B.2. Section 2 - Direct Transfer

Orbit	a $[km]$	e [-]	i $[deg]$	Ω $[deg]$	ω $[deg]$	Δt $[days]$
initial (i)	149.46×10^6	0.0160	5.27 x 10^{-3}	159.55	301.62	-
trasfer (t)	204.81×10^6	0.2813	2.37	261.05	180.07	68.24
final (f)	283.32×10^6	0.4443	4.19	298.91	168.65	-

Maneuver	previous o.	θ_{prev} $[deg]$	subsequent o.	θ_{sub} $[deg]$	Δv $[km/s]$
1 - Departure	i	340.00	t	0.05	3.969
2 - Arrival	t	69.89	f	43.51	3.226

$$\Delta v_{tot} = 7.193 \ km/s \qquad \Delta t_{tot} = 68.24 \ days$$

B.3. Section 3 - Hyperbolic Trajectories

B.3.1. Earth-Departure Hyperbola

Orbit	a $[km]$	e [-]	i $[deg]$	Ω $[deg]$	ω $[deg]$	δ $[deg]$	Δ $[km]$
parking (p)	46.750×10^3	0.7507	39.65	100.67	125.00	-	-
departure hyperbola (h)	-25.330×10^3	4.21	39.51	179.54	233.03	27.46	103.67×10^3

Maneuver	previous o.	θ_{prev} $[deg]$	subsequent o.	θ_{sub} $[deg]$	Δv $[km/s]$
H - departure	p	177.5	h	4.73	4.377

B.3.2. Asteroid-Arrival Hyperbola

Orbit	a $[km]$	e [-]	δ $[deg]$	Δ $[km]$	θ_∞ $[deg]$
arrival hyperbola (h)	$\to 0$	$\to \infty$	$\to 0$	$\to r_p$	$\to 90$
capture (c)	$\to r_p$	0	-	-	-

* where $1 \ km < r_p < 10 \ km$ is selected according to additional operational requirements.

Maneuver	previous o.	θ_{prev} $[deg]$	subsequent o.	θ_{sub} $[deg]$	Δv $[km/s]$
arrival	h	0	c	0	3.226

$$\Delta v_{patched-conic} = 7.603 \ km/s$$